

Components

- **AC-to-DC Power Adapter:** High quality AC/DC 9V 2A Switching Power Supply Adapter 2.1mm Barrel Jack
- **Barrel Connector (Female)**
- **Display:** 16x2 Serial LCD Module Display for Arduino Assembled
- **Relay:** 4-channel 5V Relay Board Module
- **Fan:** 5V Cooling Fan, 40 x 40 x 7 mm
- **DC Motor:** 3v-9v Short Shaft 130 DC Motor
- **LED:** 5V LED Traffic Light Module
- **Button:** Momentary Tactile Push Button Switch 12 x 12mm x 12mm
- **Buzzer:** Active Buzzer 5V
- **Buck:** HW-411A LM2596 DC-DC Step Down Adjustable Power Supply Module 3A Max Buck Converter
- **Current Sensor:** ACS712-5A
- **Voltage Sensor:** Voltage Detection Sensor Module 25V
- **Transistor:** BC547 NPN Transistor

Notes

- `LiquidCrystal_I2C lcd(0x27, 16, 2);`
- *Relay front* means Relay signal header; *Relay side* means Relay jumper header (here uncapped)
- *Relay has no GND on jumper header*
- *From outwards: Buck-1 IN, OUT, Buck-2 IN, OUT*

Connections

- Barrel +, Buck-1 IN +, Buck-2 IN +, ESP V_{in}
- Buck-1 OUT +, +ve rail, Current Sensor VCC, Display VCC
- Barrel -, Buck-1 IN -, Buck-1 OUT -, Buck-2 IN -, ESP GND, Current Sensor GND, Voltage Sensor -, Voltage Sensor GND, Display GND, Fan -, LED -, Motor -, Button 1,2,3 GND, GND rail
- Buck-2 OUT +, Relay side JD-VCC, Current Sensor Terminal 1
- Buck-2 OUT -, Relay front GND
- Buzzer +, +ve rail
- GPIO 14, 1k Ω
 - 1k Ω , Transistor Base
 - Transistor Emitter, GND rail
 - Transistor Collector, Buzzer -
- Display SDA, GPIO 21
- Display SCL, GPIO 22
- Relay front VCC, ESP 3.3V

- Relay side VCC — *unconnected*
- Relay IN1, GPIO 25
- Relay IN2, GPIO 26
- Relay IN3, GPIO 27
- Relay CH1 NO, LED +
- Relay CH2 NO, Fan +
- Relay CH3 NO, Motor +
- Relay CH3 COM, Relay CH2 COM, Relay CH1 COM, Current Sensor Terminal 2, Breadboard Load Rail
- Current Sensor OUT, 1k Ω
 - 1k Ω , GPIO 34
 - GPIO 34, 2k Ω
 - 2k Ω , GND
- Voltage Sensor S, GPIO 35
- Voltage Sensor VCC, Breadboard Load Rail
- Voltage Sensor + — *unconnected*
- Button 1, GPIO 18
- Button 2, GPIO 19
- Button 3, GPIO 23